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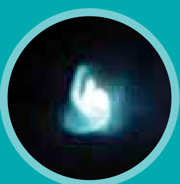
SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

The science section kicks off this month on a grim note with a reminder on how we probably have just ten years to save our planet. In other areas, we talk about a strange, inexplicable phenomenon that metals go through over time and more.



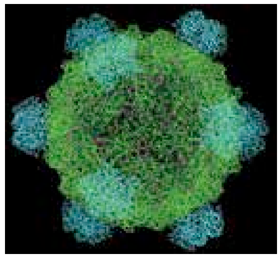
Looking back in time

Normally, black holes are formed after a star goes nova. However, theoretically, black holes can also be formed when a galaxy is born. Scientists have so far not been able to look far enough away, or far enough back in time, to observe these black holes. Researchers hope to use the James Webb Space Telescope, for this purpose.
https://dgit.in/oldbh

WHAT'S NEW

Nobel Prize in Chemistry 2018

One half of the Nobel Prize in Chemistry 2018 has been awarded to Frances H Arnold. Arnold mimicked the forces of natural selection in a lab environment, to engineer enzymes for specific purposes. Essentially, a new environmental niche is created in laboratory conditions, which encourages the existing enzymes to evolve into new ones over many generations. These enzymes, often proteins, can be used for a number of applications, including manufacturing pharmaceuticals and producing fuel. The method has even been used to create bacteria that can feed on plastics. The other half of the Nobel Prize in Chemistry 2018 is divided between George P Smith and Sir Gregory P Winter. Together, the two scientists developed methods to produce new proteins by manipulating bacteriophages – a class of microorganisms that attack bacteria. The approach developed by Smith and Sir Winter has been used to manufacture new drugs. Both the approaches are still in the early days of their use, and have a potential to tremendously benefit humans in the future.



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Nobel Prize in Physics 2018

The Nobel Prize in Physics 2018 is split in two parts, with half the prize going to Arthur Ashkin, while the other half being awarded to Gerard Mourou and Donna Strickland. Ashkin has developed optical tweezers. The laser light can be used to apply equal pressure on small objects from all sides, and keep them in the center of the beam. Atoms, molecules, and even microorganisms can be physically manipulated with these optical tweezers, without harming them. Strickland and Mourou came up with a method for generating high intensity short laser pulses. The pulses were first stretched, generated over time to reduce power consumption. Then, they were amplified, and finally they were compressed into short, high intensity pulses. The approach is used to generate the shortest, and highest intensity laser pulses ever made by man. The approach, known as chirped pulse amplification, is used for a number of applications, including eye surgeries.
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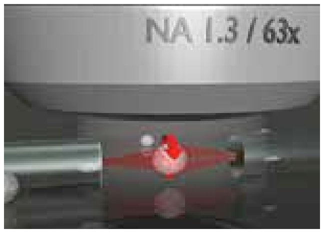
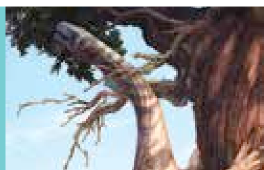
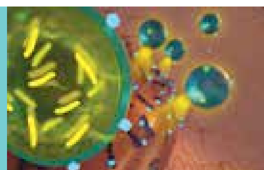




Why did dinos get so big?

Sauropods were a class of gigantic dinosaurs that were the biggest animals to ever walk the Earth. A newly discovered, well preserved fossil, from the early days of the evolution of the group, promises to show researchers exactly how and why they evolved to be so big.
https://dgit.in/brontos



Tackling drug-resistant bacteria

Researchers have developed a novel approach for combating drug resistant bacteria. A protective shell around probiotics ensures that the beneficial bacteria are not killed by antibiotics being administered. The next phase involves animal and human testing.
https://dgit.in/biotics

Helping doctors with predictions

University of Washington researchers have developed an ML system that can help surgeons predict the complications that can arise over the course of a surgery, using readings from the instruments and sensors available in a standard operation theatre.
https://dgit.in/prescience